

School mapping in ESRI imagery

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Recap from last meeting

Model fine-tuning

- All Anditi schools + equal number of OSM Vietnam non-schools
- Static Gaussian non-school sampling
- 20 epochs
- Train results
 - **F1: 99.55%**
 - P: 99.80%
 - R: 99.30%

Dense inference

- Use fine-tuned model on **overlapping satellite images** (tiles) from chosen districts
- Perform non-maximum suppression (**NMS**) to reduce the number of duplicate detections
- Measure **number of detected Anditi schools (train data)**
- Deliver **list of potential schools** sorted by prediction confidence

Dense inference results (after NMS)

- Check performance on **train data (Anditi schools)** – sanity check before evaluation
- Found 900/946 (**95.14%**) Anditi schools
- Recall: 95.14%
- 46 not detected schools (**4.86%**)
- List of potential schools: **9573 images**

Problems

- No dataset for evaluating the final fine-tuned model
- Only measured train recall, not precision and **F1 score**

New contributions

New validation datasets

- We have hand-collected two subsets of school locations:
 - Exhaustive dataset
 - Wide dataset
- Can be used for evaluating the final fine-tuned model

Exhaustive (narrow) dataset

- Locations from 4 subdistricts (wards) from **Thủ Đức District**:
 - Linh Xuan Ward
 - Binh Chieu Ward
 - Tam Binh Ward
 - Hiep Binh Phuoc Ward
- **110 schools** (exhaustive search), **1163 overlapping tiles total** (after filtering)
- Can be used for calculating precision, recall and **F1 score**

Wide dataset

- Locations from 5 districts:
 - Lê Chân District
 - Hoài Đức District
 - Cà Mau
 - Bà Rịa
 - Sóc Trăng
- **93 schools** (not exhaustive search!), **20 981 overlapping tiles total** (after filtering)
- Can be used for calculating recall

Threshold validation

- We want to determine the ideal classification threshold
- Get dense inference predictions for exhaustive dataset
- Measure F1 score on predictions for **variable thresholds (0-1)**
- Pick threshold with the **highest F1 score**
- Get recall, precision and F1 score for chosen threshold on **exhaustive dataset**

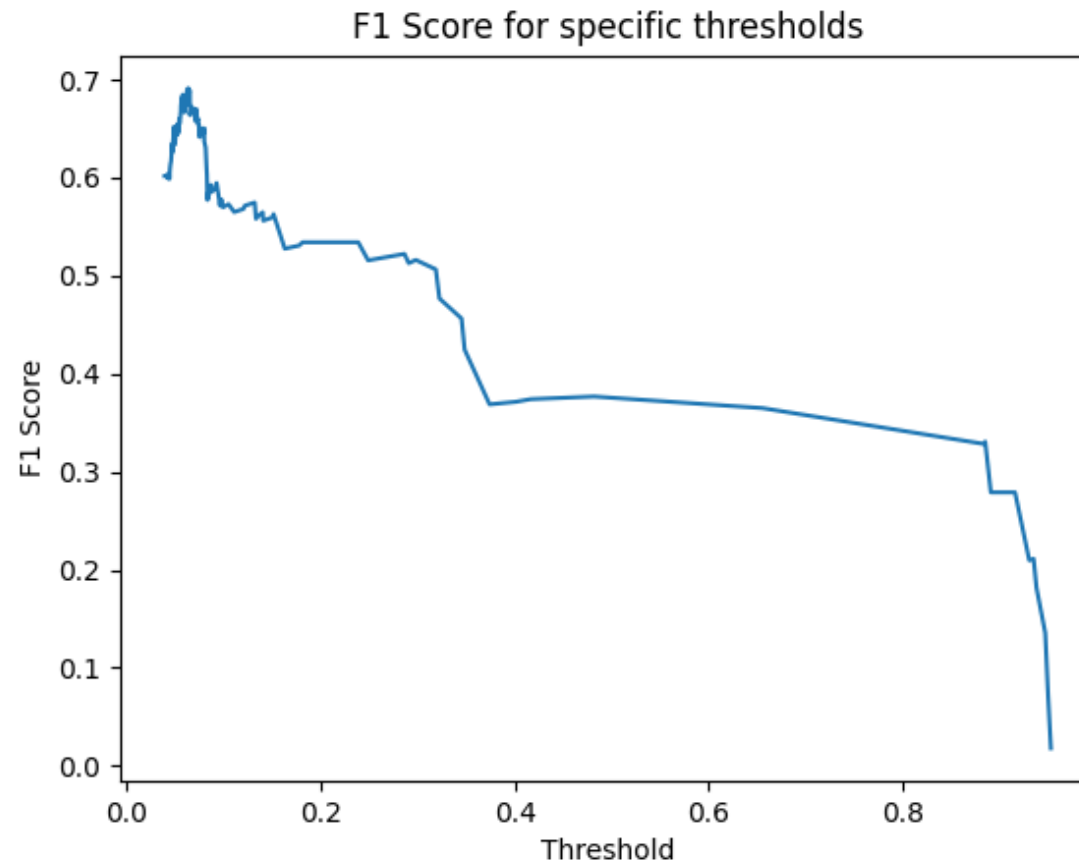
Threshold validation

- Procedure is applied to multiple models with different fine-tune configurations (fine-tune on **Anditi dataset** with non-schools sampled from **OSM Vietnam**)
- Different number of **fine-tune epochs**:
 - 1 epoch
 - 20 epochs
- **Dynamic Gaussian** non-school sampling with different school : non-school sampling ratios:
 - 1:1
 - 1:10

Threshold validation, results

School : non-school ratio	Fine-tune epochs	Best threshold (%)	Best F1 score (%)	Best Recall (%)	Best precision (%)
1:1	20	3.61	61.85	97.27	45.34
1:10	20	4.04	63.70	84.55	51.10
1:10	1	6.33	69.14	76.36	63.16

Threshold validation, 1:10 ratio, 1 epoch



Conclusion & Future work

- F1 score for fine-tuned model (1:10 ratio, 1 epoch):
 - Train: 88.38%
 - **Exhaustive: 69.14%**
- Fine-tuning for 20 epochs may overfit the nation-wide model to Anditi dataset
- Explore and test options for reducing overfitting